

# Logical & Physical Data Independence

**Q.** Define Data Independence. Explain the concepts of Logical and Physical Data Independence and highlight the key differences between them.

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> **Definition to Remember**

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## 1. Types of Data Independence

Data independence is linked to the three-level abstraction architecture and is classified into two types:

#### 1. Physical Data Independence

#### 2. Logical Data Independence

## 2. Difference: Physical vs Logical Data Independence

| Feature    | Physical Data Independence                                                                                                                     | Logical Data Independence                                                                                                                         |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition | Physical data independence is the ability to change the physical storage of data without affecting the logical structure of the data.          | Logical data independence is the ability to change the logical structure of the data without affecting the physical storage of the data.          |
| Example    | Changing the physical storage of data from one disk to another.                                                                                | Changing the logical structure of data from one schema to another.                                                                                |
| Impact     | Physical data independence is concerned with the physical storage of data and is not affected by changes in the logical structure of the data. | Logical data independence is concerned with the logical structure of the data and is not affected by changes in the physical storage of the data. |

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> **Must-Write Points (for 10 marks)**

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> **Quick Recall**

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